Computer Science & IT

Theory of Computation



CFL

Lecture No. 10



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Recap of Previous Lecture



- Topic
- Topic
- Topic

LL(k) LR(k)

Topics to be Covered



- Topic
- Topic
- Topic

LR, LF, PYQ

Removal of left recursion :-

$A \Rightarrow A\alpha$

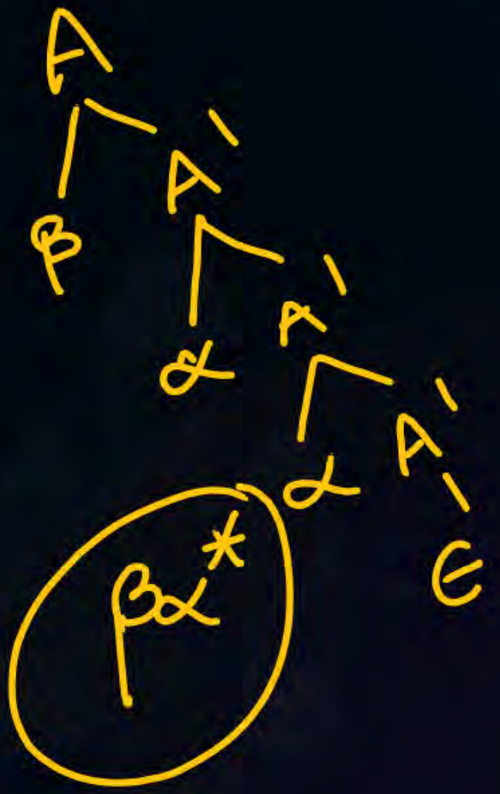
$A \rightarrow A\alpha / \beta$

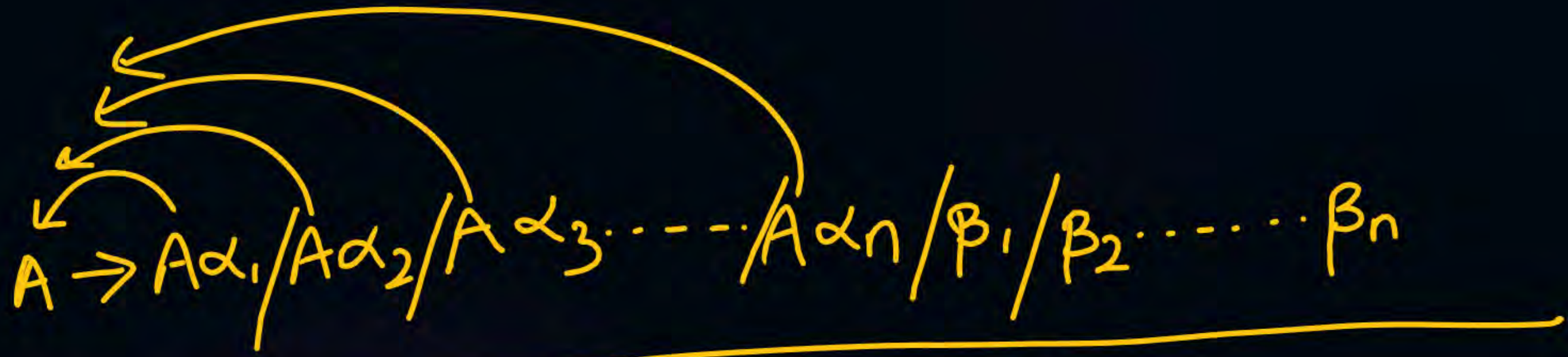
$A \Rightarrow \beta\alpha^*$

$A \rightarrow \beta A'$

$A' \rightarrow \alpha A' / \epsilon$







$A \rightarrow \beta_1 A' / \beta_2 A' / \beta_3 A' \ldots \beta_n A'$

$A' \rightarrow \alpha_1 A' / \alpha_2 A' / \alpha_3 A' / \ldots \alpha_n A' / \epsilon$

$$S \rightarrow S\underset{\alpha_1}{+}S \,/\, S\underset{\alpha_2}{*}S \,/\, \underset{\beta_1}{a} \,/\, \underset{\beta_2}{b} \,/\, \underset{\beta_3}{c}$$

$$S \rightarrow aS' \,/\, bS' \,/\, cS'$$

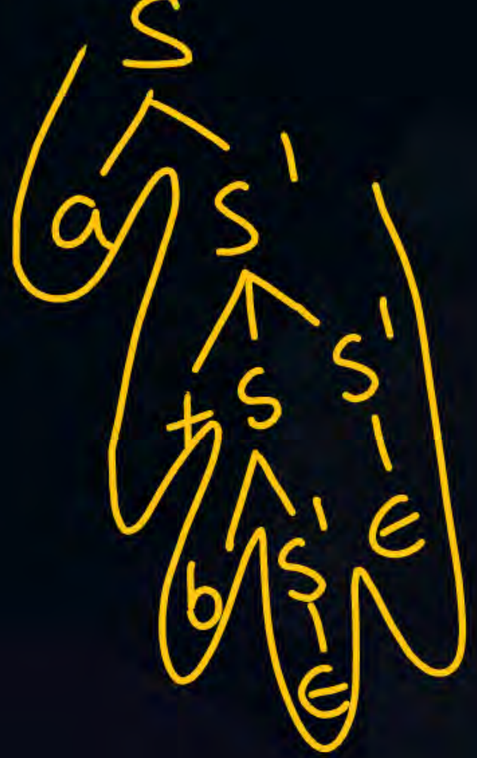
$$S' \rightarrow +SS' \,/\, *SS' \,/\, \epsilon$$

$S \rightarrow S+S / S*S / a / b / c / \epsilon$

$\alpha_1 \quad \alpha_2 \quad \beta_1 \quad \beta_2 \quad \beta_3 \quad \beta_4$

$S \rightarrow aS' / bS' / cS' / \epsilon S'$ → S'

$S' \rightarrow +SS' / *SS' / \epsilon$



Removing left factoring:-

$$A \rightarrow \alpha x / \alpha y / \alpha z$$

$$A \rightarrow \alpha A'$$

$$A' \rightarrow x / y / z$$

$S \rightarrow aA\epsilon / aAb / aAB / B$

$S \rightarrow aAA' / B$

$A' \rightarrow \epsilon / b / B$

Left factoring and left recursion can be eliminated,
But it need not eliminate ambiguity
Removing ambiguity is undecidable.

PYQs:-

A CFG is ambiguous if

(a) the grammar contains useless non terminals

(b) It produces more than one parse tree for some sentence

(c) Some production has two non terminals side by side

(d) Non of the above

→ CFLs and Regular languages are closed under the operation(s) of

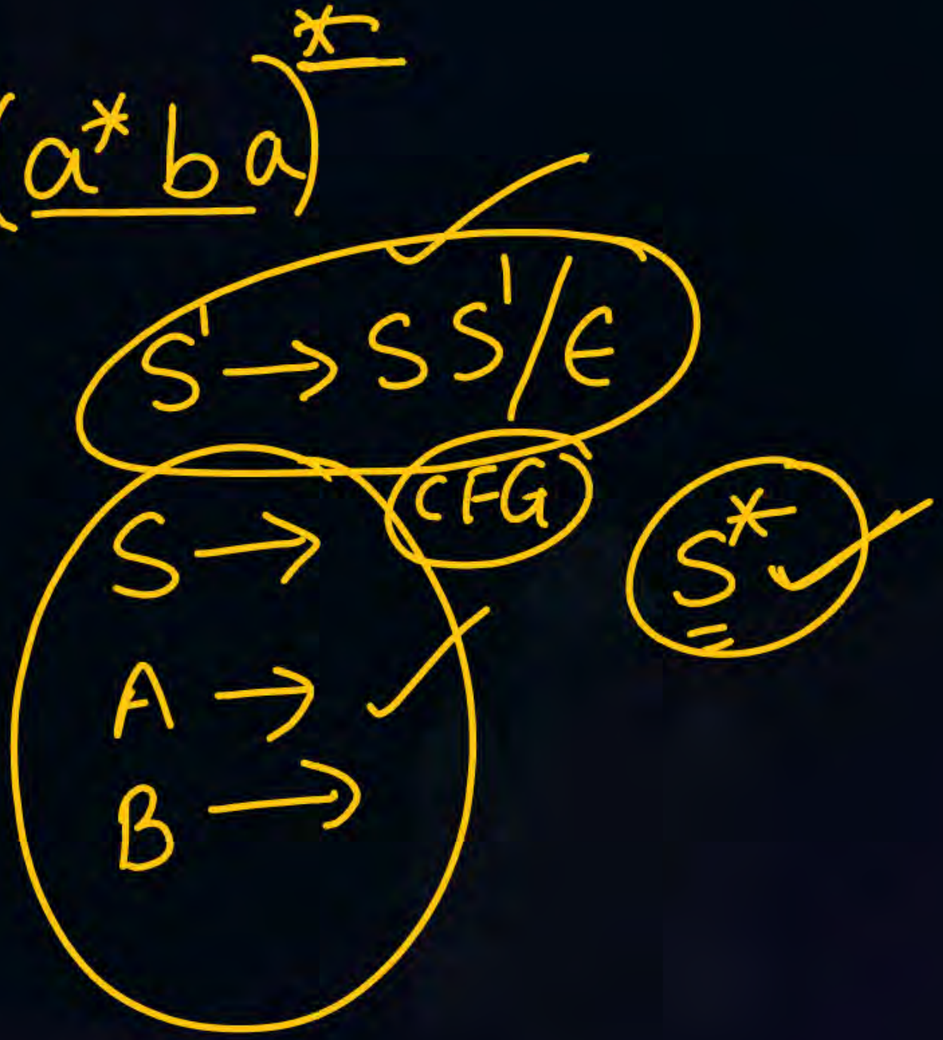
a) union b) intersection

c) concatenation d) complementation

Context free languages are

a) closed under union
b) closed under complementation
c) closed under intersection
d) closed under Kleen closure

RL = $(a^* b a)^*$



$L \cap R$ ✗ CFL

Table 2

CFL ∩ CFL → ~~SL~~

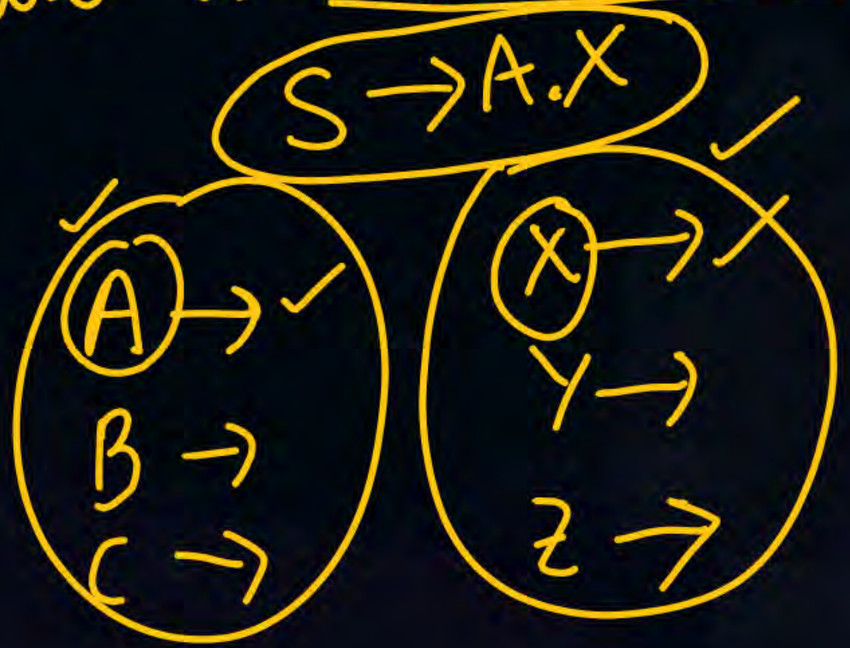
→ If L_1 and L_2 are CFLs and R is a regular set one of the languages below is not necessarily a CFL. which one?

a) $L_1 \cdot L_2$ — CFL

b) $L_1 \cap L_2$

c) $L \cap R$ — CFL

d) $L_1 \cup L_2$ — CFL



$L \cap R$ ✓ ✓ ✗ CFL

CFL ∩ CFL → ✗ SL

Table 2

→ If L_1 and L_2 are CFLs and R is a regular set one of the languages below : a CFL. which one?

a) $L_1 \cdot L_2$ CFL ✓

b) $L_1 \cap L_2$ ✗ ✓

c) $L \cap R$ CFL

d) $L_1 \cup L_2$ CFL ✓

$S \rightarrow A / X$ ✓

A → ✓

B →

C →

X → ✗

Y →

Z →

→ If G is a CFG, and w is a string of length 'n' in L(G), how long is the derivation of w in G, if G is a CNF?

a) 2n b) 2n+1 c) 2n-1 d) n

$S \to AB$

$A \to a$

$B \to b$

$|ab| = 2$ ✓

$(2(2)-1)$

$S \overset{1}{\Rightarrow} AB \overset{2}{\Rightarrow} aB \overset{3}{\Rightarrow} ab$

→ let G be a CFG with following productions

$S \to ABA/c$

$A \to aA/\epsilon$

$B \to bB/\epsilon$

$C \to d$

nullable = {A, B}

eliminate ϵ productions

→ which of the following language is accepted by DPDA

a) $\{wcw^R / w \in \{a,b\}^*\}$ → DPDA

b) $\{ww^R / w \in \{a,b\}^*\}$ → NPDA

c) $\{a^n b^n c^n / n \geq 0\}$ → CSL

d) $\{w / w$ is a palindrome over $\{a,b\}\}$ → NPDA

$a^n b^n \subseteq (a+b)^*$

$a^n b^n$ — CFL

$(a+b)^*$ — Regular

$a^n b^n \subseteq a^* b^*$

Which of the following statement is false?

a) Every finite subset of a non regular set is regular

b) Every subset of a regular set is regular

c) Every finite subset of a regular set is regular

d) Intersection of two regular sets is regular

$a^n b^n \subseteq (a+b)^*$

CFL are closed under

a) Union, ~~intersection~~

b) union, Kleene closure

c) ~~intersection~~, ~~complement~~

d) ~~complement~~, Kleene closure

→ If L_1 is CFL and L_2 is a regular language which of the following is/are false

Xa) $L_1 - L_2$ is not CFL

b) $L_1 \cap L_2$ is context free

Xc) $\bar{L}_1$ is context free

d) $\bar{L}_2$ is regular

$L_1 \cap \bar{L}_2$

CFL ∩ Reg = CFL ✓

Reg → DFA → comp

The language accepted by a PDA in which the stack is limited to 10 items is best described as

(a) Context free
~~(b) Regular~~
(c) DCFL
(d) recursive → HTM



$a^n b^n$

$a^{10} b^{10}$

FA

FA → Finite memory → States

G:

$S \to aSb / SS / \epsilon$ ✓

$S \to (S) / SS / \epsilon$

Perfectly balanced parantheses

which of the following is true

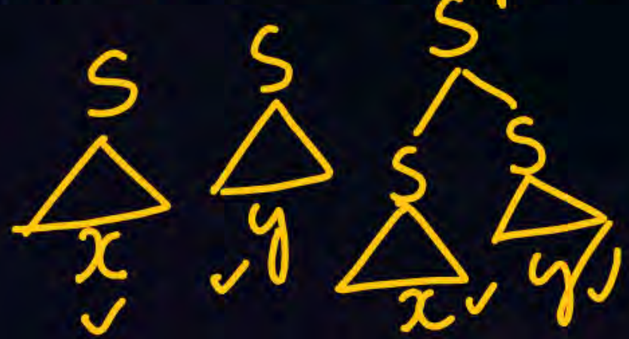
~~a) G is not ambiguous~~

~~b) There exists $x, y \in L(G)$ such that $xy \notin L(G)$~~

c) There is a DPDA that accepts L(G)

✗d) There is a DFA that accepts L(G)







→ Which of the following is false

a) There exists CFL's such that all CFGs generating them are ambiguous (inherently ambiguous languages)

b) An unambiguous CFG always has a unique parse tree for each string of the language generated by it

X c) Both DPDA and NPDA accepts the same set of languages

d) A finite set of strings from some alphabet is always a regular language.

$a^m b^n c^{m+n} / n, m \geq 1$

Pu Pu PdP

a) Regular

b) Context free but not regular

c) Context sensitive but not context free

d) Type 0 but not Context sensitive

→ Let L be a Regular language and m be a Context free language. Let L^c and m^c denotes complements of L and m respectively

What is $L^c \cup m^c$

$\overline{Reg} \cup \overline{CFL} = Reg \cup CSL = CSL$ ✓

~~a)~~ It is necessarily regular but not necessarily context free

~~b)~~ It is necessarily context free

~~c)~~ It is necessarily non regular

~~d)~~ Non of the above



In the context free grammar given below, S is start symbol a and b are terminals and ϵ denotes the empty string

$S \to aSa/bSb/a/b/\epsilon$ → palindromes

which of the following strings is not generated by the grammar

a) aaaa b) baba c) abba d) babaaaabab

Let $L_1 = \{ 0^{n+m} 1^n 0^m / n, m \geq 0 \}$ → CFL (DCFL)

$L_2 = \{ 0^{m+n} 1^{m+n} 0^m / n, m \geq 0 \}$ → $0^x 1^x 0^y$ $(x > y)$ X CFL (CSL)

$L_3 = \{ 0^{n+m} 1^{n+m} 0^{n+m} / n, m \geq 0 \}$ $0^x 1^x 0^x$ ∉ CFL ∴ X CFL, CSL

Which of the languages are not Context free

a) L_1 only b) L_3 only c) L_1 and L_2

d) L_2 and L_3

THANK - YOU